

Week-2 Module

Q1. Fraud Transaction Detection (Linear Search)

Problem Statement :-

A bank stores transaction ID's in the order they occur. Since the transactions are not sorted, the security team needs to check if a suspicious transaction ID exists. Write program using linear search.

Program :-

```

import java.util.*;
public class Main {
    public static void main( String [] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int [] arr = new int[n];
        for (int i=0; i<n; i++) {
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        boolean found = false;
        for (int i=0; i<n; i++) {
            if (arr[i] == target) {
                found = true;
                break;
            }
        }
        if (found) System.out.println("Fraud Transaction found");
        else System.out.println("Transaction Not found");
    }
}

```

Q2 Server log search (Binary search)

Problem statement:

A cloud company stores server log ID's in sorted order. Engineers need to quickly verify if a log ID exists. Write a program using Binary search.

Sol

Program :-

```
import java.util.*;
public class Main {
    public static void main (String [] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int [] arr = new int [n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        int l = 0, r = n - 1;
        boolean found = false;
        while (l <= r) {
            int mid = (l + r) / 2;
            if (arr[mid] == target) {
                found = true;
                break;
            }
            else if (arr[mid] < target)
                l = mid + 1;
            else r = mid - 1;
        }
        if (found) system.out.println("log found");
        else system.out.println("log not found");
    }
}
```

Q3

Sort Employee salaries (Bubble Sort)

Problem Statement:

A company wants to sort employee salaries in ascending order for payroll analysis. Write a program using bubble sort.

sol

Program :-

```

import java.util.*;
public class Main () {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        int [] arr = new int [n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for (int i = 0; i < n-1; i++) {
            for (int j = 0; j < n-i-1; j++) {
                if (arr[j] > arr[j+1]) {
                    int temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp;
                }
            }
        }
        for (int x : arr) System.out.print (x + " ");
    }
}

```


Q4

Sort Website Response Time (Insertion Sort)

Problem Statement :

A web monitoring system records website response times. New data comes continuously and must be inserted into the correct position. Write program using Insertion Sort.

Sol

Program :

```
import java.util.*;
public class Main {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        int [] arr = new int [n];
        for (int i=0; i<n; i++) {
            arr [i] = sc.nextInt();
        }
        for (int i=1; i<n; i++) {
            int key = arr [i];
            int j = i-1;
            while (j>=0 && arr [j] > key) {
                arr [j+1] = arr [j];
                j--;
            }
            arr [j+1] = key;
        }
        for (int x: arr) System.out.print (x + " ");
    }
}
```

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Prioritize bug reports (selection sort)

Problem statement:

A software company assigns priority scores to bug reports. To resolve bugs efficiently, they want to sort priorities in ascending order. Write program using selection sort.

Sol,

Program:-

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = sc.nextInt();
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for (int i = 0; i < n - 1; i++) {
            int minIdx = i;
            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[minIdx]) {
                    minIdx = j;
                }
            }
            int temp = arr[i];
            arr[i] = arr[minIdx];
            arr[minIdx] = temp;
        }
        for (int x : arr) System.out.print(x + " ");
    }
}
```